

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

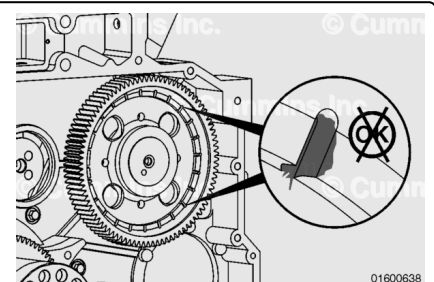
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html>).
- Remove the left-bank camshaft or right-bank idler gear. Refer to Procedure 001-037 (</qs3/pubsys2/xml/en/procedures/56/56-001-037-tr.html>) or Refer to Procedure 001-038 in Section 1 (</qs3/pubsys2/xml/en/procedures/56/56-001-038-tr.html>).
- Remove the accessory idler gear.



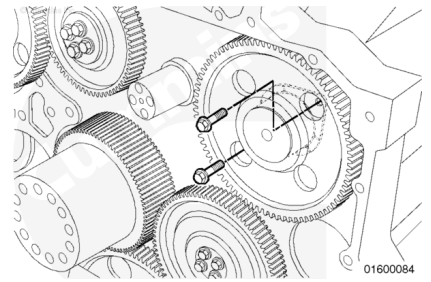
Initial Check

- Make sure the target notches are free of debris or metal burrs.



Remove

Remove the two capscrews from the camshaft thrust bearing.



Mechanical Gear Remover

Assemble the Mechanical Gear Remover, Part Number 3162895.

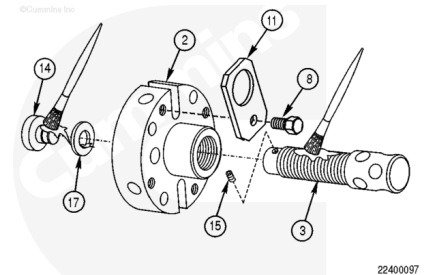
Apply anti-seize or heavy duty grease onto the threads of the remover screw (3).

Install the remover screw (3) into the remover plate assembly (2).

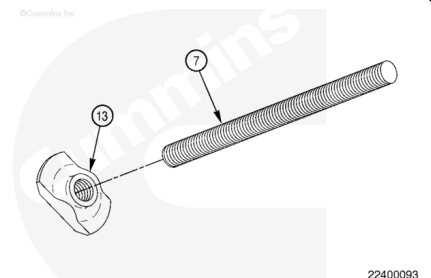
Lubricate the thrust washer (17) with anti-seize or heavy duty grease.

Assemble the thrust washer (17) and remover screw adapter (14) to the remover screw (3), and secure it with the socket setscrew (15).

Install plate (11) on the remover plate assembly (2) with one capscrew (8) and tighten capscrew.



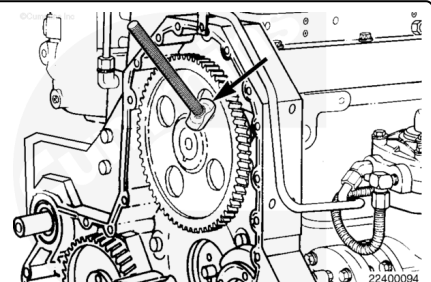
Install the remover jaw (13) onto the end of the stud (7). The end of the stud **must** be flush with the bottom of the jaw.



Note : The round blanking holes in the camshaft gear must be aligned in the 1:00 and 7:00 positions to insert the remover jaw assembly.

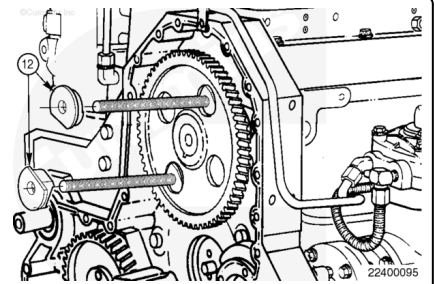
Tilt the remover jaw and stud assembly through a round blanking hole, and maneuver it sideways until the remover jaw is inserted.

Install the other remover jaw and stud assembly in the opposite round blanking hole.



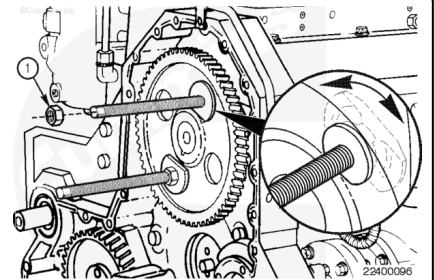
Position the flat face of the retainer flange (12) toward the camshaft.

Slide the retainer (12) down the studs until it is positioned into the camshaft gear blanking hole and engages with the remover jaw.



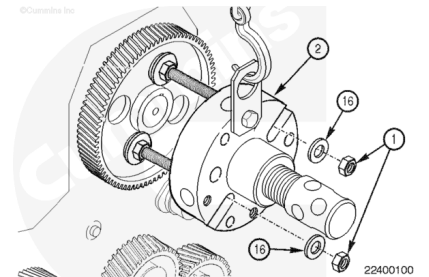
Note : Make certain that the flat face of the retainer flange sits against the radius of the camshaft gear hub. This will locate the remover jaw correctly at the back of the camshaft gear.

Adjust the retainer if necessary, and uniformly hand tighten the nuts (1) onto the studs.



⚠ WARNING ⚠

The remover is heavy and the gear can come loose suddenly causing the gear and gear remover assembly to fall. To reduce the possibility of personal injury and component damage, use a hoist or other lifting device when using the gear remover.



Using a lifting device, slide the remover plate (2) over the studs and position it close to the camshaft gear.

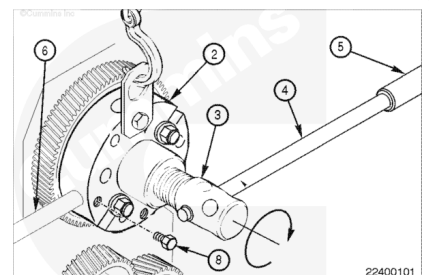
Make sure the pilot of the old remover screw adapter is properly engaged with the camshaft.

Install the washers (16) and nuts (1) onto the studs.

Uniformly hand tighten the nuts.

⚠ CAUTION ⚠

Do not attempt to alter the camshaft gear remover assembly for use with any kind of impact wrench. Do not exceed 890 N [200 lb] of force on the end of the handle. The threads on the remover can gall and seize.



Make sure anti-sieze or heavy duty grease is applied to the remover screw threads (3).

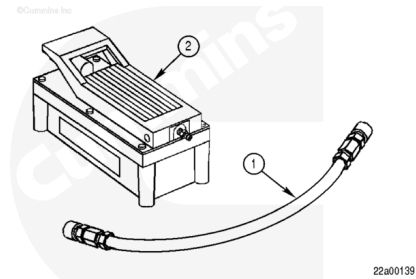
Install the holding handle (6) into the remover plate assembly (2) and secure it with one capscrew (8).

Using handles (4) and (5), and holding handle (6), turn the remover screw (3) **clockwise** to remove the camshaft gear from the camshaft.

Hydraulic Gear Remover

These items are used with the hydraulic gear remover kit, Part Number 3164375, and one or more of the engine-specific kits.

Item	Part Number	Description	Quantity
1	3823624	Hydraulic hose	1
2	3823622	Air/hydraulic pump	1

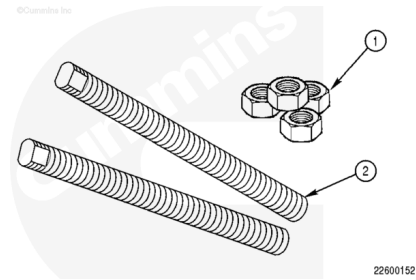


A minimum of 621 kPa [90 psi] of clean, dry shop air supply is required for use of this tool.

Check the fluid level in the air/hydraulic pump.

As an alternative to kit, Part Number 3164378, four studs and four nuts can be purchased individually, if **only** the capability of removing the camshaft gear is required.

Item	Part Number	Description	Quantity
1	3279355	Nut, M16 x 2	4
2	3163558	Stud, M16 x 2 (2 shown)	4



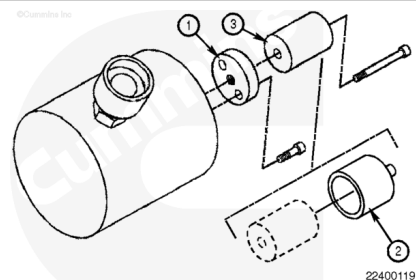
Note : Use spacer base (1) where necessary.

Place the spacer base (1), Part Number 3163766, into the center of the 30-ton hydraulic cylinder, Part Number 3163551. Secure it with two socket head capscrews.

Attach spacer (3), Part Number 3164201, to the spacer base with the socket head capscrew, Part Number 3163446.

Spacer (3) is used for other engine applications.

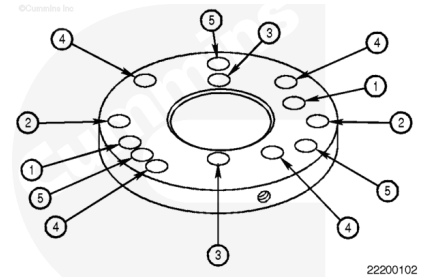
Check the hydraulic cylinder for leaks and any signs of damage.



The remover plate is designed to be used on several different engine applications. The mounting holes in the remover plate, Part Number 3163553, have been marked for identification.

1. L10, M11, ISM, and QSM11

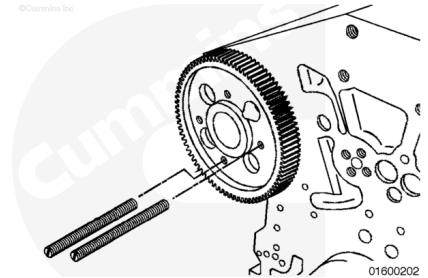
2. K19, K38, and K50
3. QST30 and QSK19
4. QSK45 and QSK60
5. QSK23.



22200102

Note : When installing the studs, make certain that the end of the stud is flush with the back of the gear.

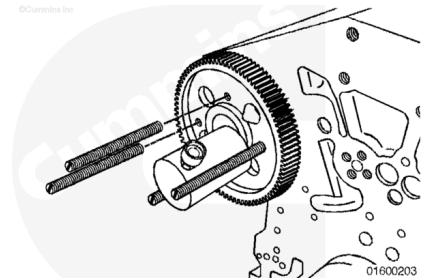
Install two studs into the threaded holes of the camshaft gear as illustrated.



Note : The spacer, Part Number 3164201, and the socket head capscrew, Part Number 3163446, must be removed from the 30-ton hydraulic cylinder.

Place the 30-ton hydraulic cylinder over the two studs.

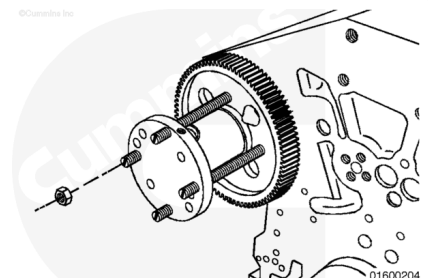
Install the remaining two studs into the camshaft gear.



Note : Make certain that the 30-ton hydraulic cylinder is located in the counterbore provided in the remover plate.

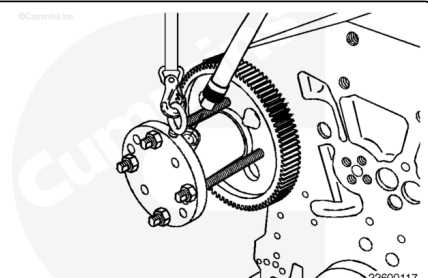
Install the remover plate over the four studs and against the 30-ton hydraulic cylinder. Make certain that the 30-ton hydraulic cylinder assembly is centered on the camshaft and seated in the counterbore of the remover plate.

Install and uniformly hand-tighten the four nuts onto the studs.



⚠ WARNING ⚠

The gear remover is heavy and the gear can come loose suddenly, causing the gear and gear remover assembly to fall. To reduce the possibility of personal injury and component damage, use a hoist or other lifting device when using the gear remover.



Install the eyebolt into the remover plate.

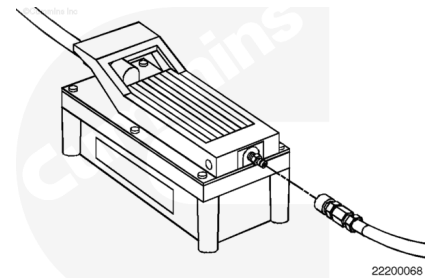
Attach a lifting device to the eyebolt. Lift the camshaft gear remover kit slightly, if necessary, to align it with the camshaft.

⚠ WARNING ⚠

To reduce the possibility of personal injury, remain clear of pressurized hoses, which can rupture suddenly or break free.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

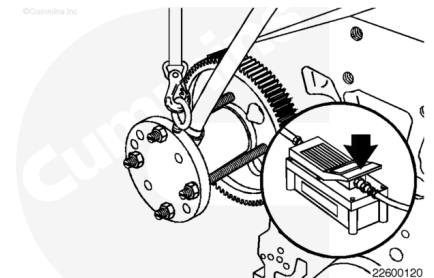


Connect the air hose to the input side of the air/hydraulic pump. Connect the hydraulic hose to the output side of the pump.

Connect the other end of the hydraulic hose to the 30-ton hydraulic cylinder.

⚠ CAUTION ⚠

Make certain that the 30-ton hydraulic cylinder is located in the counterbore of the remover plate and the piston is centered on the camshaft when it makes contact. Failure to comply can damage the gear, camshaft, or camshaft gear remover kit.



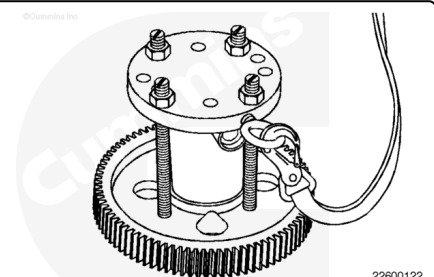
Slowly operate the air/hydraulic pump until the gear is removed.

Remove the hydraulic camshaft gear remover kit and the camshaft gear as one unit.

Note : To reduce the possibility of damage to the 30-ton hydraulic cylinder, release the hydraulic pressure. This will retract the piston.

Remove the hydraulic hose.

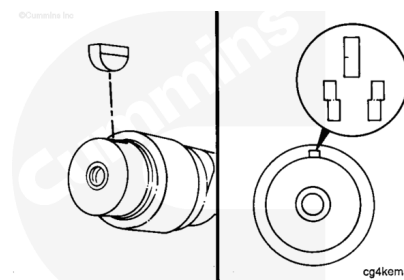
Disassemble the hydraulic camshaft gear remover from the camshaft gear.



Note : Engines with electronically actuated injectors do not use offset keys.

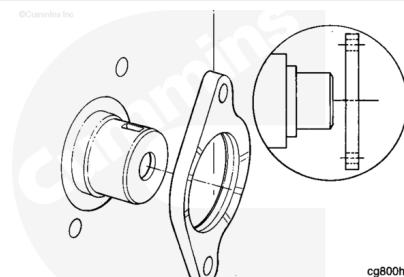
Check and record the direction of the key offset when compared to the camshaft gear rotation (same or opposite).

Remove the key.



Mark the position of the camshaft thrust bearing before removing it.

Remove the camshaft thrust bearing.



Magnetic Crack Inspect

⚠ CAUTION ⚠

Use a copper braid contact that has neoprene covers to reduce the possibility of burning the teeth of the gear.

If the gear contains a keyway, position the gear so that the keyway is pointing toward one of the contacts.

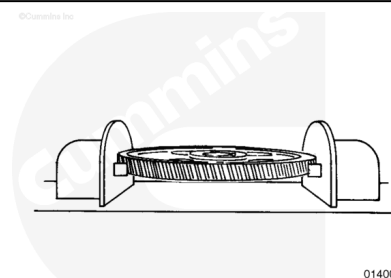
Use a magnetic particle testing machine.

Use the residual method and apply head shot amperage.

Adjust the amperage to the value specified in the table.

Amperage for Gear Sizes	
Gear Outside Diameter	Direct Current Amperage
Less than 101 mm [4 in]	1000
101 to 203 mm [4 to 8 in]	1500
Greater than 203 mm [8 in]	2000

Check the camshaft gear for cracks.



If the gear contains a keyway, position the gear so that the keyway is pointing toward the coil.

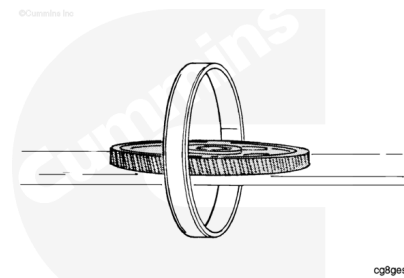
Use the residual method and apply coil shot amperage.

Adjust the amperage to the value specified in the table.

Amperage for Gear Sizes	
Gear Outside Diameter	Ampere Turns
Less than 101 mm [4 in]	4000
101 to 203 mm [4 to 8 in]	6000
Greater than 203 mm [8 in]	8000

Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Check the camshaft gear for cracks.



⚠ CAUTION ⚠

The gear must be demagnetized completely and cleaned thoroughly. Any small particles will cause engine damage.

An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface is visible with the use of the ultraviolet light.

Do **not** use the gear if:

Limits of Acceptance - Machined Surfaces

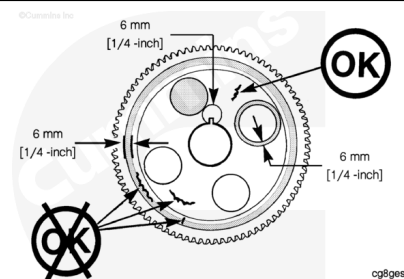
- There is an open indication.
- There is an indication below the surface that is in the shaded area, shown in the graphic.
- There is an indication below the surface that is longer than 6 mm [0.250 in].

Limits of Acceptance - Forged Surfaces

- There is a open indication that is in a circumferential direction.
- There is an open indication that is longer than 9.5 mm [0.375 in].
- There is an indication below the surface that is in the shaded area shown in the graphic.

An indication below the surface in a radial direction is acceptable if it is in the area that is **not** shaded.

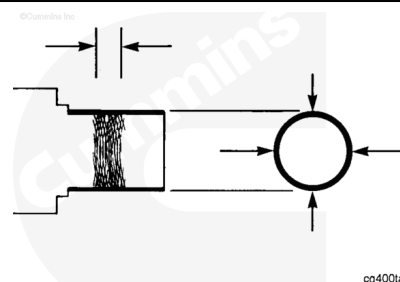
Demagnetize the gear.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the camshaft and gear with solvent and dry with compressed air.

Check the camshaft for fretting damage. The camshaft **must** be replaced if the fretting damage is more than 3 mm [0.125 in] wide.

Measure the outside diameter of the camshaft.

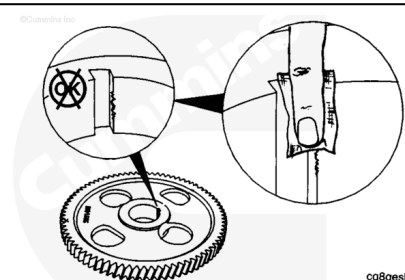
Camshaft Outside Diameter (Gear Location)		
mm		in
86.90	MIN	3.421
86.99	MAX	3.425

If the camshaft is **not** within specifications. The camshaft **must** be replaced.

Check the keyway of the camshaft gear for burrs.

Remove any burrs with a fine emery cloth.

If the burrs can **not** be removed with a fine emery cloth or, the keyway is damaged the camshaft gear **must** be replaced.



Inspect the bore of the gear for frets or burrs.

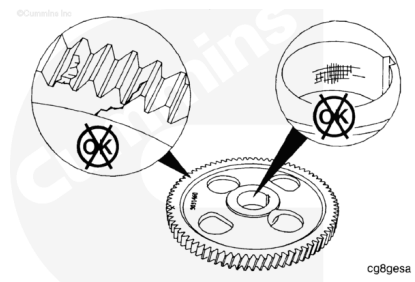
If fretting is in excess of 3 mm [0.125 in] wide, the camshaft gear **must** be replaced.

If the fretting or burrs can **not** be removed with emery cloth, the camshaft gear **must** be replaced.

Measure the inside diameter of the camshaft gear.

Camshaft Gear Inside Diameter		
mm		in
86.88	MIN	3.420
86.90	MAX	3.421

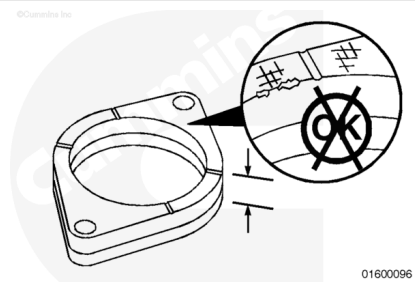
If the camshaft gear is **not** within specifications, it **must** be replaced.



Measure the camshaft thrust bearing thickness.

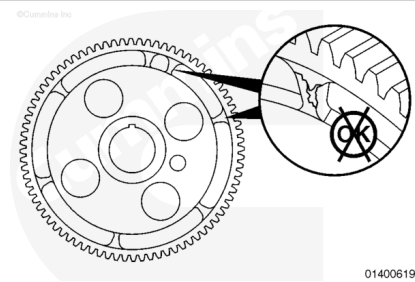
Camshaft Thrust Bearing Thickness		
mm		in
13.95	MIN	0.549
14.05	MAX	0.553

If the camshaft thrust bearing is **not** within specifications, it **must** be replaced.



with Electronically Actuated Injector

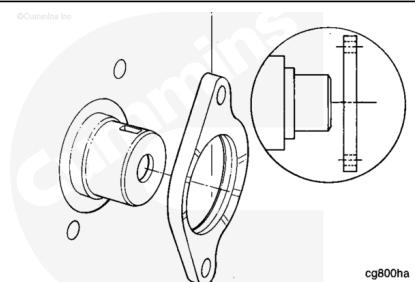
Inspect the backside of the gear for cracks, chipped, or broken bosses that the cam position sensor reads. The gear **must** be replaced if the bosses are damaged.



Install

The gear housing **must not** prevent the thrust plate from being flat against the block.

Use Lubriplate™ 105, Part Number 3163086 to lubricate the outside diameter of the camshaft at the gear location.



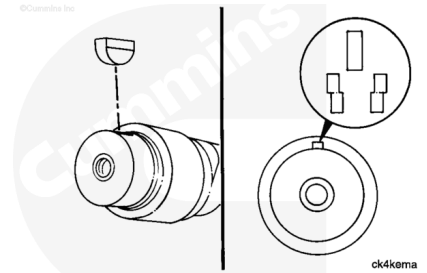
Install the thrust plate with the grooves in the thrust plate positioned toward the camshaft gear.

Note : Engines with electronically actuated injectors do not use offset keys.

Camshaft keys are available in different sizes (amount of offset). The injection timing is controlled by:

- The selection of the camshaft key
- The direction of the key offset when compared to the camshaft gear rotation
- The amount of offset.

Install the same part number key in the same orientation as the key that was removed. Refer to Procedure 006-025 in Section 6 (/qs3/pubsys2/xml/en/procedures/56/56-006-025-tr.html).



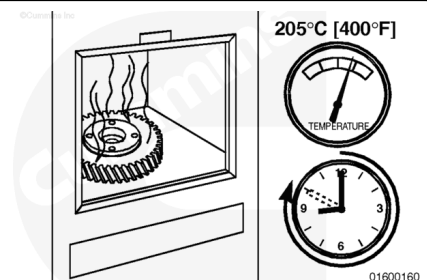
⚠ CAUTION ⚠

Do not exceed the specified time or temperature. Damage to the gear and the gear teeth will result.

⚠ CAUTION ⚠

To reduce the possibility of damage to the camshaft gear, do not attempt to install the gear without using heat.

Use an oven and adjust the heat to 205°C [400°F]. Heat the gear for a minimum of 40 minutes and a maximum of 1 hour. The inside diameter of the gear will become larger and simplify installation.

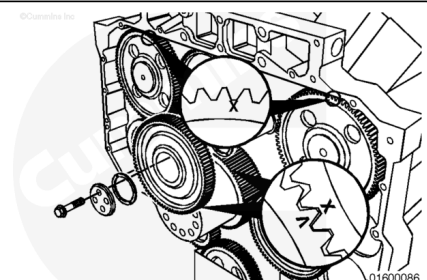


⚠ WARNING ⚠

To reduce the possibility of severe burns, wear protective gloves when installing the heated camshaft gear.

⚠ CAUTION ⚠

The camshaft must be held in its most forward position when the camshaft gear is being installed. Failure to hold the camshaft in this position can result in improper location of the camshaft gear on the camshaft.



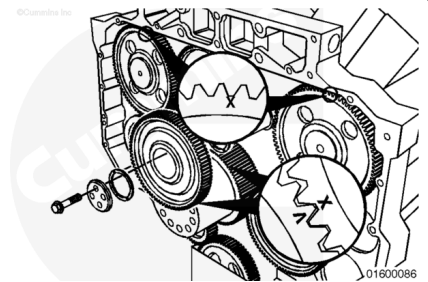
Remove the gear from the oven, and install it on the camshaft.

The keyway in the gear **must** be aligned with the key in the camshaft.

⚠ CAUTION ⚠

Allow the gear to cool slowly. Do not use water or oil to reduce the cooling time. This can cause the gear to crack.

The timing mark on the camshaft gear **must** be visible from the front of the gear after it is installed on the camshaft.

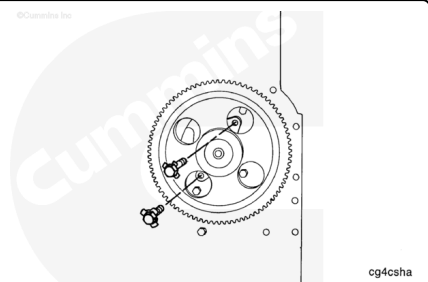


If necessary, rotate the crankshaft so that the holes in the camshaft gear allow access to the capscrew holes in the thrust plate.

Install the two thrust plate mounting capscrews.

Tighten the capscrews.

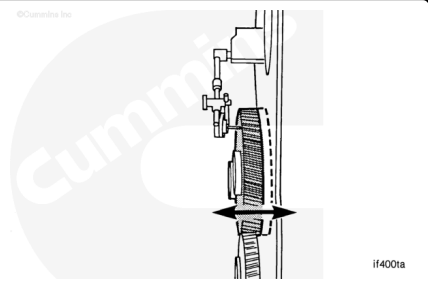
Torque Value: 45 n•m [33 ft-lb]



Use a dial indicator, Part Number 3376050 or equivalent, to measure the camshaft end clearance.

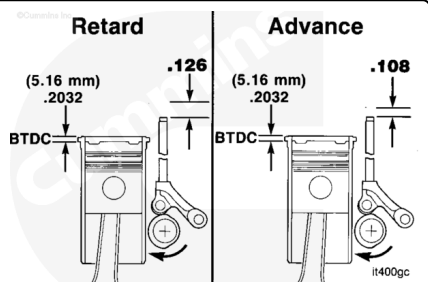
Camshaft End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013

If the camshaft end clearance is **not** within specification, check for foreign material between the thrust plate and the block, and proper location of the gear on the camshaft.



If the camshaft gear was replaced, check the injection timing; machine tolerances can cause timing to be out of specifications. Refer to Procedure 006-025 in Section 6 (/qs3/pubsys2/xml/en/procedures/56/56-006-025.html).

Install the camshaft idler gear(s). Make sure all timing marks are correctly matched.



Finishing Steps

- Install the left-bank camshaft or right-bank idler gear.
Refer to Procedure 001-037
(</qs3/pubsys2/xml/en/procedures/56/56-001-037-tr.html>) or Refer to Procedure 001-038 in Section 1
(</qs3/pubsys2/xml/en/procedures/56/56-001-038-tr.html>).
- Install the accessory idler gear.
- Install the front gear cover and related components. Refer to Procedure 001-031 in Section 1
(</qs3/pubsys2/xml/en/procedures/56/56-001-031.html>).
- Operate the engine and check for leaks.

